

Question: How far should we rely on medical science to attain good health?

Key words:

How far – answer should address the importance of EXTENT: large or limited.

should -- idea of desirability

rely on -- depend on

medical science – encompasses the practice of the diagnosis, treatment and prevention of disease/illness (whether of body or of mind) and would include medication, health supplements, medical devices and surgery. Students need to evaluate the characteristics of medical science (reliability, efficacy and impact).

good health – this encompasses the physical/mental/emotional/psychological aspects of health.

Possible stand:

It is more defensible to argue that we should rely on medical science to a **limited extent**.

Suggested Approach:

Part A: Medical science has been perceived to be a modern-day panacea for all matters of ill health and most rely on it extensively/ to a large extent to maintain good health.

Part B: Concede that while medical science has led to improvements in health for many, the belief that medical science should be relied on for good health to a large extent is actually an illusion as its effects are limited. Students should evaluate the limitations of medical science followed by the evaluation of other factors which affect one's health as well.

Part A: Reasons why there are people who believe that we should rely on medical science to a large/significant extent to maintain good health:

1. Arguably, such strong reliance on medical science is a result of previous successes of medical science in reducing, eradicating a medical condition or even in preventing further deterioration through vaccinations and/or medications.

- Most people rely on medical science to a large extent to enjoy good health because of medical advances that have been achieved and comprehensive health systems being improved continually to deal with health issues and to increase longevity for many.

- A recent example would include the serious threat the Ebola virus posed from 2014 to 2016. With more than 25,000 cases of Ebola in West Africa and 11,325 deaths from the deadly virus during 2014-2016, patients were heavily reliant on medical science to prevent fatality. Infected individuals often had to be rehydrated with oral or intravenous fluids - and treatment of specific symptoms improved survival. In addition, patients were given the rVSV-ZEBOV vaccine in the 2018-2019 Ebola outbreak in the DRC. The two-dose regimen of Ad26.ZEBOV and MVA-BN-Filo, sold under the brand names Zabdeno and Mvabea was developed by Johnson & Johnson.

- Other notable successes in the past were the eradication of smallpox and significant reductions of serious infections like polio and measles.

-The last reported case of smallpox occurred in Somalia in 1977. Since then, health officials have scaled back eradication campaigns and no new infections have been detected, leading to declarations of eradication for smallpox in 1980 by the World Health Organization through the extensive use of vaccinations.

-Following the widespread use of poliovirus vaccine in the mid-1950s, the incidence of poliomyelitis declined rapidly in many industrialized countries. Czechoslovakia became the first country in the world to scientifically demonstrate nationwide eradication of poliomyelitis in 1960. Nigeria is the latest country to have officially stopped endemic transmission of wild poliovirus, with its last reported case in 2016.

2. People also pin their hopes on medical science to a large extent in the matter of good health because of the ongoing advancements such as the concept of personalised medicine or pharmacogenomics which promises better results.

-This concept of personalised medicine is based on the understanding that different individuals have different responses to drugs due to different genes. Hence, matching the right drugs to patients not only has obvious clinical benefits but reduces the occurrence of misdiagnosis, long hospitalisation periods and the need for long follow-up periods which have rendered medical science problematic in some ways thus far.

-But this approach of detecting and diagnosing diseases in a highly specific manner, categorizing them by genetic variation rather than by symptom, promises greater success for patients and their families, increasing their confidence in medical science to a large extent to restore good health. In short, the reliability of medical science to target the root cause of a medical problem has led to many trusting it to a large extent.

For example, genomic analysis can reveal genetic variations which can guide drug dosage, meaning that patients can get maximal drug benefit with minimal side effects. This has been applied to a whole range of diseases so far, such as cancer, mental illness and inflammatory bowel disease.

Additionally, melanomas, a type of skin cancer that develops from the pigment-producing cells known as melanocytes, typically occur in the skin but may rarely occur in the mouth, intestines or eye (uveal melanoma). Typical treatment for melanomas are not viable in these patients, and therefore personalised treatment dramatically increases prognosis (chances of recovery).

3. Even people who are healthy depend on health supplements to a large extent and consume vitamins on a daily basis to maintain good health.

- This is possibly because they are even more aware today of the health risks present in modern lifestyles and wish to prevent the onset of disease.

- Also, given the reality of aging populations and the need to stay in the workforce for a longer period of time, many feel the need to invest in their long-term health to be able to perform well economically to secure longer periods of employment.

-Examples: Biochemist Bruce Ames from the University of California, Berkeley, has developed a list of 41 vitamins and minerals such as magnesium and omega-3 fatty acids that he says may help with healthy aging.

Also, many people consume probiotics for good gut health and to decrease inflammation.

Part B: Reasons why there are others who believe that we should rely on medical science only to a limited extent to maintain good health:

4. While it is undeniable that medical science has helped many people to recover and maintain good health, it is also a fact that unintended side effects can lower the quality of life which suggests that heavy reliance on medical science/medicine is undesirable.

-This is often because an integral part of healthcare is based on risk-benefit analysis which weighs the potential for side effects against the potential for positive outcomes of a given treatment.

-For example, in the treatment for diabetes, the use of drugs like metformin can produce a dangerous side effect called lactic acidosis – a serious condition where the cells of the body do not get enough oxygen to survive. It is caused by a build-up of lactic acid in the blood, especially for people whose kidneys are not working well which is often the case for diabetic people. Yet, metformin is commonly administered in the treatment of diabetes based on the principle that the benefits outweigh the risks.

-2nd example: According to the Foundation for a Drug-Free World, short-term side effects of anti-depressants include slowed brain function, slowed breathing, lowered blood pressure, poor concentration, confusion, and sluggishness. Patients have also been examined to be disoriented, dizzy and serious cases even had seizures.

5. Medical science often fails to address the root causes of poor health and instead addresses the symptoms of an illness. Hence, they often recur and make undue reliance on medical science questionable.

-Root causes often need attention to a host of factors such as changes in lifestyle, diet and lowering of stress levels which medical science is unable to redress adequately as they are often tied to the individual's willingness to make lifestyle changes.

-This means medication alone is ineffective in ensuring good health for the individual and hence, it is unrealistic to expect medical science to solve all health-related problems without addressing the other factors.

Evidence: The pragmatic nature of modern society is largely responsible for poor lifestyle choices such as sedentary lifestyles, fast-food options, high stress levels and inadequate sleep. But to make dramatic changes to such a lifestyle is considered a luxury which most blue-collared workers cannot afford as they need to provide for their families.

- Example: According to researchers from the Finnish Institute for Health and Welfare, lifestyle choices often affect the life expectancy of men and women. The biggest causes for shortened life expectancy for typical 30-year-old men are smoking and diabetes. Smoking takes 6.6 years and diabetes 6.5 years out of their life expectancy. Being under heavy stress also shortens

their life expectancy by 2.8 years. In such cases, medical science is limited in lengthening an individual's lifespan and restoring health.

6. Medical science is limited in its capability to resolve all health problems as it is playing catch-up most of the time. Thus, heavy reliance on medical science is inadvisable.

- Medical science responds to the emergence of health problems and tries to provide answers as best it can but cannot completely help with a range of health problems such as autoimmune diseases, mental health issues, ADHD or dyslexia; it can only help to alleviate them.
- This is because medical science is always trying to increase its depth and breadth of knowledge of complex bodily processes and can never fully claim to have perfect answers to existing and emerging medical problems.

Example 1: According to the Professor and Director of the Infectious and Inflammatory Disease Centre, a cure for autoimmune disease is much more challenging. Unlike cancer or infection, where the goal is to get rid of bad cells in the body, we must maintain an immune system. The key is to reduce inflammation by resetting the controls and help tissues and organs regenerate from the damage caused by inflammation.

Example 2: It is important to remember that even with optimal treatment and adherence, a good deal of the impairment that results from ADHD is not effectively ameliorated. This stems in part from the underlying pathophysiology of the disorder which is both complex and yet poorly understood. Another mediator is the limitation of current interventions to repair the disordered brain processes that are constitutive of ADHD. As with many other neurodevelopmental disorders, treatment is primarily symptom-focused rather than curative.